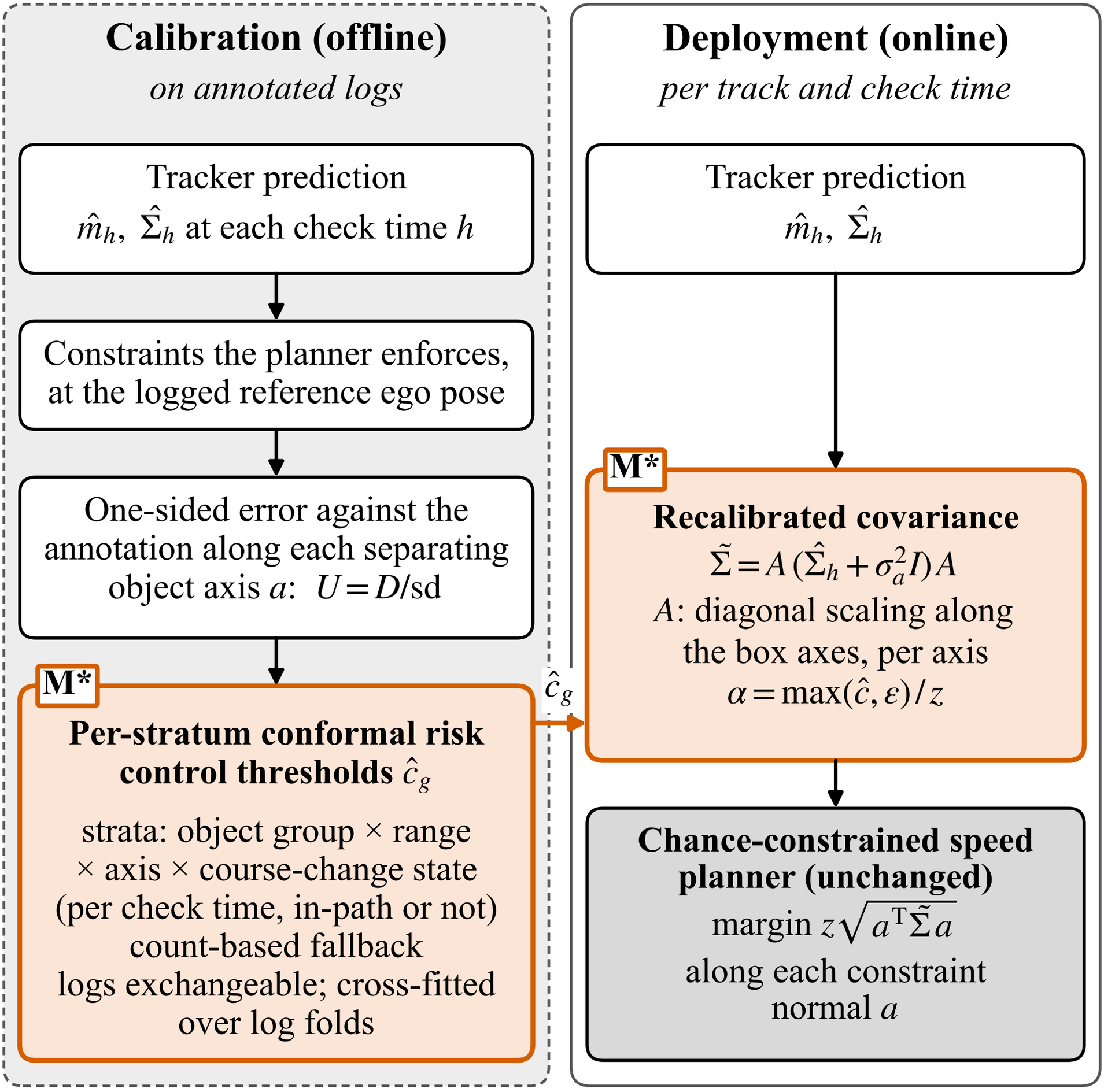

Calibration (offline)
on annotated logs
Tracker prediction
$\hat{m}_h$, $\hat{\Sigma}_h$ at each check time h
Constraints the planner enforces, at the logged reference ego pose
One-sided error against the annotation along each separating object axis a: $U = D/\mathrm{sd}$
M*
Per-stratum conformal risk control thresholds $\hat{c}_g$
strata: object group × range × axis × course-change state (per check time, in-path or not)
count-based fallback
logs exchangeable; cross-fitted over log folds
$\hat{c}_g$
Deployment (online)
per track and check time
Tracker prediction
$\hat{m}_h$, $\hat{\Sigma}_h$
M*
Recalibrated covariance
$\tilde{\Sigma} = A\,(\hat{\Sigma}_h + \sigma_a^2 I)\,A$
A: diagonal scaling along the box axes, per axis
$\alpha = \max(\hat{c}, \varepsilon)/z$
Chance-constrained speed planner (unchanged)
margin $z\sqrt{a^{\mathrm{T}}\tilde{\Sigma}a}$
along each constraint normal a